Code: 20A02705 **R20**

B.Tech IV Year I Semester (R20) Supplementary Examinations April/May 2025

RENEWABLE ENERGY SYSTEMS

(Common to CE, ME, ECE, CSE, IT, FT, AI&DS, CSE (AI), CSE (DS), CSE (AI&ML), CSE (IOT), CSE (CS), CS&D and AI&ML)

Time: 3 hours Max. Marks: 70

PART – A

(Compulsory Question)

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1 Answer the following: (10 X 02 = 20 Marks)

(a) What are the key components of a flat plate solar collector? 2M
(b) Explain the concept of thermal storage in solar energy systems. 2M
(c) How does temperature affect the performance of a PV cell? 2M
(d) What are the key factors that influence the efficiency of a solar PV module? 2M
(e) What are the differences between horizontal and vertical axis wind turbines? 2M
(f) What factors influence the selection of a wind energy site? 2M
(g) How is geothermal energy generated inside the earth crust? In India where is geothermal energy available? 2M
(h) What is the geothermal gradient, and why is it important? 2M
(i) Describe how wave energy can be converted into electrical energy? 2M
(j) Discuss the various methods to improve the kinetic performance of a fuel cell. 2M

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

2 (a) Briefly explain extraterrestrial Solar Radiation with relevant diagrams. 5M
(b) Briefly discuss any two ways of solar energy storage methods. 5M

OR

3 (a) Explain how local solar time is calculated. What is the importance of local solar time in solar energy systems? 5M
(b) A small household-lighting system is powered from a nominally 8 V (i.e. four cells at 2 V) storage battery having a 30 Ah supply when charged. The lighting is used for 4.0 h each night at 3.0 A. Design a suitable photovoltaic power system that will charge the battery from an arrangement of Si solar cells. 5M

(i) How will you arrange the cells?
(ii) How will the circuit be connected?
How will you test the circuit and performance?

4 (a) Discuss the design considerations for PV systems in remote locations. What factors determine the feasibility and reliability of such systems? 5M
(b) Explain how grid-connected PV systems operate? Discuss the role of inverters, and the challenges of integrating solar power into the grid. 5M

OR

5 (a) Describe the environmental and economic benefits of using PV systems for remote and grid-connected power generation. 5M
(b) A PV system costs $5000 to install and has an annual energy output of 3000 kWh. If the cost of electricity is $ 0.12 per kWh, calculate the energy payback time for the system. 5M

6 (a) Explain Maximum Power Point Tracking procedure for a Wind System. 5M
(b) Describe the working principle of a wind energy conversion system (WECS). What are the key components, and how do they function together to convert wind energy into electrical energy? 5M

OR

Contd. In Page 2

R20

7	(a)	A wind turbine has a rotor diameter of 60 meters and operates at a site with an average wind speed of 8 m/s. Calculate the power available in the wind. Assume the air density is 1.225 kg/m^3.	5M
	(b)	Describe the aerodynamic forces that act on wind turbine blades, including lift and drag. How do these forces affect the efficiency and performance of the wind turbine?	5M
8	(a)	Explain the different types of geothermal resources: hydrothermal, geo-pressured, hot dry rocks, and magma.	5M
	(b)	At a geothermal site, the temperature at the surface is 30°C, and at a depth of 1500 m, the temperature is 180°C. Calculate the geothermal gradient at this location.	5M
		OR	
9	(a)	How is geothermal energy used for direct applications? Give examples of geothermal energy being used for district heating, greenhouse farming, or industrial processes.	5M
	(b)	A geothermal reservoir has a volume of 2×10^6 m^3 with an average temperature of 200°C. If the specific heat of the reservoir rock is 850 J/kg°C and the density is 2500 kg/m^3, calculate the total thermal energy stored in the reservoir.	5M
10	(a)	Explain the working and constructional details of a biogas generation plant. What factors should be considered when selecting a site for the plant and designing the digester?	5M
	(b)	Discuss different types of fuel cells what are the advantages of fuel cell energy.	5M
		OR	
11	(a)	Discuss briefly the operation of solid oxide fuel cell (SOFC).	5M
	(b)	Explain the working principle of a tidal energy power plant. Discuss its advantages and limitations compared to other renewable energy technologies.	5M

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Code: 20A02705 **R20**

B.Tech IV Year I Semester (R20) Regular & Supplementary Examinations December/January 2024/2025

RENEWABLE ENERGY SYSTEMS

(Common to CE, ME, ECE, CSE, IT, FT, AI&DS, CSE (AI), CSE(DS), CSE (AI&ML),CSE(IOT), CSE(CS) CS&D and AI&ML)

Time: 3 hours Max. Marks: 70

PART – A

(Compulsory Question)

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1 Answer the following: (10 X 02 = 20 Marks)

(a)	Differentiate between beam and diffuse solar radiation.	2M
(b)	Explain what earth-sun angles are and how they affect solar energy availability.	2M
(c)	List the basic components of a photovoltaic (PV) system.	2M
(d)	What are the advantages of thin-film PV cells over crystalline silicon cells?	2M
(e)	How is wind energy converted into electrical energy in a wind turbine?	2M
(f)	What factors influence the selection of a wind energy site?	2M
(g)	What are the limitations of harnessing Geo-thermal energy?	2M
(h)	How is electricity generated from Geo-thermal energy?	2M
(i)	List the different biomass conversion technologies.	2M
(j)	State the working principle of a proton exchange membrane (PEM) fuel cell.	2M

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

2 (a) Calculate the time taken for the water at 40^oC in a cooking pot to boil in a solar cooker with the following specifications: 5M

$(\alpha\tau) = 0.7$; $F' = 0.85$, $U_L=6\ W/m^2\ ^oC$, $A_p = 0.36\ m^2$,

$MC = 4 \times 4190\ J/^oC$; $Ta=15\ ^oC$, $T_{wo} = 40^oC$, $T_w = 100^oC$

I_T (with + without reflector) = (400+600)¼ 1000 $W/m^{2.}$

(b) Explain the concept of solar constant and how it is determined? Why is it considered a fundamental parameter in solar energy studies? 5M

OR

3 (a) Explain how local solar time is calculated? What is the importance of local solar time in solar energy systems? 5M

(b) How are sunrise, sunset, and day length calculated for a specific location? Discuss how these factors impact the performance of solar energy systems? 5M

4 (a) A PV system costs $5000 to install and has an annual energy output of 3000 kWh. If the cost of electricity is $0.12 per kWh, calculate the energy payback time for the system. 5M

(b) What are the challenges and limitations associated with the large-scale deployment of PV systems, both in remote areas and grid-connected environments? 5M

OR

5 (a) What are the electrical characteristics of silicon PV cells and modules? Discuss parameters like open-circuit voltage, short-circuit current, fill factor, and efficiency. 5M

(b) Explain the principle of the photovoltaic effect in crystalline silicon cells. How does the structure of the cell enable the conversion of sunlight into electricity? 5M

Contd. in Page 2

6	(a)	What are the advantages of horizontal axis wind turbines (HAWT) over vertical axis wind turbines (VAWT)?	5M
	(b)	Describe the aerodynamic forces that act on wind turbine blades, including lift and drag. How do these forces affect the efficiency and performance of the wind turbine?	5M
		OR	
7	(a)	A wind turbine with a 50-meter rotor diameter operates with an efficiency of 40% at a location where the wind speed averages 10 m/s. Calculate the annual energy production in MWh, assuming the turbine operates 3000 hours per year.	5M
	(b)	Describe the working principle of a wind energy conversion system (WECS). What are the key components, and how do they function together to convert wind energy into electrical energy?	5M
8	(a)	How is geothermal energy used for direct applications? Give examples of geothermal energy being used for district heating, greenhouse farming, or industrial processes.	5M
	(b)	A geothermal reservoir has a volume of 2 x 10^3 m^3 with an average temperature of 200°C. If the specific heat of the reservoir rock is 850 J/kg°C and the density is 2500 kg/m^3, calculate the total thermal energy stored in the reservoir.	5M
		OR	
9	(a)	Compare the working of a flash steam geothermal power plant with a binary cycle power plant.	5M
	(b)	Discuss the economic aspects of geothermal energy projects. What are the initial costs, operational costs, and the return on investment?	5M
10	(a)	Discuss the various technologies used for harnessing wave energy. Compare their performance in terms of efficiency and adaptability to different ocean conditions.	5M
	(b)	A biomass power plant uses 1000 kg of dry wood per hour as fuel. If the calorific value of the wood is 18 MJ/kg and the plant operates with an efficiency of 30%, calculate the electric power output of the plant.	5M
		OR	
11	(a)	Compare the performance of different fuel cell technologies based on efficiency, operating temperature, and fuel requirements.	5M
	(b)	What are the environmental and economic benefits of biomass energy? Highlight its role in reducing carbon emissions and supporting rural economies.	5M

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Code: 20A02705

R20

B.Tech IV Year I Semester (R20) Supplementary Examinations July/August 2024

RENEWABLE ENERGY SYSTEMS

(Common to CE, ME, ECE, CSE, IT, FT, AI&DS, CSE (AI), CSE (DS), CSE (AI&ML) and CSE (IOT))

Time: 3 hours | Max. Marks: 70

PART – A

(Compulsory Question)

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1		Answer the following: (10 X 02 = 20 Marks)	
	(a)	State various types of single axis tracking solar thermal collectors.	2M
	(b)	Which equipment is used to measure the global and diffused solar radiations?	2M
	(c)	Draw the equivalent circuit of solar PV cell.	2M
	(d)	State the various losses taking place in Solar PV Panels.	2M
	(e)	Why is taper and twist provided to horizontal axis wind turbine?	2M
	(f)	State the difference between stall controlled and pitched controlled horizontal axis wind turbines.	2M
	(g)	State the various places in India where geothermal energy is available.	2M
	(h)	What are the barriers in harnessing geothermal energy?	2M
	(i)	State the advantages and applications of Biomass.	2M
	(j)	State the various types of fuel cells.	2M

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

2 State the classification of solar thermal collectors. Explain with neat sketch compound parabolic thermal collector. 10M

OR

3 State different types of thermal energy storage systems and discuss in detail hybrid thermal energy storage system. 10M

4 Discuss solar PV cell on following points (i) Importance of space charge region, (ii) PV cell manufacturing technologies and cell efficiencies, (iii) I-V characteristics, (iv) P-V characteristics and (v) Fill factor. 10M

OR

5 Discuss grid connected solar PV system on following points (i) Block diagram of the system, (ii) MPPT control and (iii) Control of grid connected inverter. 10M

6 Explain horizontal axis pitch-controlled wind turbine on the following points (i) Pitch angle control below and above the rated speed, (ii) Power output of the turbine with respect to wind speed and (iii) Aerodynamic breaking mechanism. 10M

OR

7 Discuss the various components of horizontal axis wind turbine and state the advantages and disadvantages of horizontal axis wind turbine. 10M

8 Discuss the types of geothermal resources and explain with block diagram electrical power generation with geothermal energy. 10M

OR

9 Discuss geothermal energy on the following points (i) Effects on the environment, (ii) sustainability and (iii) Economics related to harnessing the energy. 10M

10 Discuss tidal power (electricity) generation methods and state the advantages and disadvantages of tidal energy. 10M

OR

11 Discuss alkaline fuel cell on following points, (i) Construction, (ii) Operation with chemical reactions and (iii) Applications. 10M

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Code: 20A02705

R20

B.Tech IV Year I Semester (R20) Regular Examinations December/January 2024

RENEWABLE ENERGY SYSTEMS

(Common to CE, ME, ECE, CSE, IT, FT, AI&DS, CSE(AI), CSE(DS), CSE(AI&ML) & CSE(IOT)

Time: 3 hours | Max. Marks: 70

PART – A

(Compulsory Question)

*****

1		Answer the following: (10 X 02 = 20 Marks)	
	(a)	State various types of stationary solar thermal collectors.	2M
	(b)	Which equipment is used to measure beam (direct) solar radiations?	2M
	(c)	Why grid connected solar PV systems are more popular as compared off grid solar PV systems?	2M
	(d)	What are the causes of Hot Spot in Solar PV module?	2M
	(e)	Why is gearbox required in wind turbine power generation system?	2M
	(f)	State the advantages and disadvantages of horizontal axis wind turbine.	2M
	(g)	State the types of Geothermal resources.	2M
	(h)	State the applications of Geothermal energy.	2M
	(i)	State the advantages and disadvantages of fuel cell.	2M
	(j)	State the advantages and disadvantages of tidal energy.	2M

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

2	Compare single axis tracking solar thermal collector with dual axis tracking solar thermal collector. Discuss with neat sketch.	10M
	OR	
3	Define and discuss following terms associated with solar radiations (i) Air mass (ii) local apparent time, (iii) angle of incidence and (iv) declination angle.	10M
4	Discuss various losses taking place in the Solar Photo Voltaic panel. Draw the set of I-V and P-V characteristics for the range of radiations varying from 200 W/m^2 to 1000 W/m^2.	10M
	OR	
5	Explain the equivalent circuit of the solar PV cell and discuss the effect on I-V characteristics with change in series and shunt resistance. State the importance of blocking and bypass diode in series parallel connection of solar PV cells.	10M
6	Derive the expression for turbine power output for horizontal axis wind turbine. Discuss turbine output power control below and above rated speed.	10M
	OR	
7	Discuss following terms associated with horizontal axis wind turbine (i) aero foil, (ii) angle of attack (iii) twist and taper of turbine blade, (iv) aerodynamic brakes and (v) mechanical brakes.	10M
8	State the advantages and disadvantages of geothermal energy and discuss sustainability and barriers associated with geothermal energy.	10M
	OR	
9	Discuss various types of power plants used to harness electrical power from geothermal energy.	10M
10	State the ideal properties of electrode and electrolyte required for fuel cell. Discuss phosphoric acid fuel cell on following points, (i) construction, (ii) operation with chemical reactions and (iii) applications.	10M
	OR	
11	Discuss various process of converting biomass into electricity.	10M

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Code: 20A02705s **R20**

B.Tech IV Year I Semester (R20) Supplementary for MOOCs (Conventional mode) Examinations June 2025

RENEWABLE ENERGY SYSTEMS / SOLAR ENERGY ENGINEERING & TECHNOLOGY

Time: 3 hours Max. Marks: 70

PART – A

(Compulsory Question)

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1		Answer the following: (10 X 02 = 20 Marks)	
	(a)	What is meant by terrestrial radiation?	2M
	(b)	Define solar constant.	2M
	(c)	Define the short circuit current of the PV module.	2M
	(d)	What is solar cell?	2M
	(e)	State Bouger's law.	2M
	(f)	Define absorptivity.	2M
	(g)	What is the function of the absorber plate in a solar air heater?	2M
	(h)	Name two types of solar thermal power generation systems.	2M
	(i)	What is sensible heat storage?	2M
	(j)	Mention any one passive architectural design strategy for solar cooling.	2M

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

2	(a)	Discuss about Extraterrestrial and Terrestrial solar radiation.	5M
	(b)	Write the applications of solar energy.	5M
		OR	
3		Discuss about the instruments used to measure the solar radiation.	10M
4		Explain in detail the various configurations of the stand alone PV system.	10M
		OR	
5	(a)	What are the different Standard PV module parameters? Discuss all of the parameters.	5M
	(b)	Discuss all the factors affecting electricity generated by a solar cell.	5M
6		Enumerate the different types of solar collectors with necessary diagram.	10M
		OR	
7		Explain the working principle of a liquid flat plate solar collector with a neat diagram. Discuss the role of each component.	10M
8		Describe the standard testing procedure of a solar air heater.	10M
		OR	
9		Discuss the working principle and construction of any two types of solar thermal power generation systems.	10M
10	(a)	Briefly explain about solar distillation.	5M
	(b)	Discuss about the emerging technologies in solar systems.	5M
		OR	
11		Explain in detail about the types of thermal energy storage.	10M

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Code: 20A02705s **R20**

B.Tech IV Year I Semester (R20) Advanced Supplementary for MOOCs (Conventional mode) Examinations July 2024

RENEWABLE ENERGY SYSTEMS

(For 2020 Regular & 2021 Lateral Entry admitted batches only)

Time: 3 hours Max. Marks: 70

PART – A

(Compulsory Question)

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1 Answer the following: (10 X 02 = 20 Marks)

	(a)	Define solar constant. What is its standard value?	2M
	(b)	Differentiate terrestrial and extra-terrestrial solar radiation.	2M
	(c)	What are the advantages of concentrating collectors?	2M
	(d)	What is meant by biomass fuel?	2M
	(e)	Discuss economic aspects of biogas.	2M
	(f)	What are the techniques suggested for maintaining bio-gas production?	2M
	(g)	Classify different wind turbine rotors.	2M
	(h)	Why horizontal axis wind turbines are preferred over vertical axis wind turbines?	2M
	(i)	Write short notes on the wind energy pattern factor.	2M
	(j)	What are the relative features of drag and lift type machines in Windmills.	2M

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

2		What are the reasons for variation in solar radiation reaching the earth and that received outside the earth atmosphere?	10M
		OR	
3		Explain following angles used in solar radiation analysis: (i) Latitude of location, (ii) Hour angle, (iii) Solar azimuth angle, (iv) Zenith angle, (v) Declination angle.	10M
4		Describe the application of solar energy photovoltaic energy conversion. What are the limitations? Suggest steps to overcome its demerits.	10M
		OR	
5		Explain in detail the working principal and thermal analysis of flat plate collectors.	10M
6		What are types of thermal energy storage systems? Draw and explain briefly about thermal energy storage systems.	10M
		OR	
7		With a neat sketch, explain the working of solar pond electric power plant.	10M
8		Write a short note on bio-gas plant.	10M
		OR	
9		Explain briefly about Pyrolysis process.	10M
10	(a)	Explain the operation wind energy system with a neat sketch.	5M
	(b)	Show that ideal maximum power coefficient is 0.59 for a horizontal axis windmill.	5M
		OR	
11	(a)	Explain different types and characteristics of windmill rotors with relevant diagrams.	5M
	(b)	Discuss the merits and demerits associated with wind energy systems.	5M

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